ModelIt!

Teacher's Guide

The Ocean's Silent Emergency

Modeling Ocean Acidification, pH Tipping Points, and Marine Ecosystem Collapse

Quick Reference

Grade Level	Grade 11L2
NGSS Standard	HS-ESS2-6, HS-LS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-6, HS-LS2-6

Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere. Evaluate claims, evidence, and reasoning that the complex interactions in ecosystems maintain relatively consistent numbers and types of organisms in stable conditions, but changing conditions may result in a new ecosystem.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a quantitative model of ocean carbon chemistry to predict how rising CO_2 concentrations alter pH, carbonate availability, and marine ecosystem stability.
Disciplinary Core Idea	**ESS2.D Weather and Climate / LS2.C Ecosystem Dynamics** — Carbon cycling between atmosphere and hydrosphere drives ocean acidification, which disrupts ecosystem dynamics by changing the chemical conditions organisms depend on for survival.
Crosscutting Concept	**Stability and Change / Cause and Effect** — Students analyze how small, steady changes in atmospheric CO_2 can push ocean chemistry past critical thresholds, triggering large-scale ecosystem instability through cascading cause-and-effect chains.

CAST Testing Alignment

MULTIPLE CHOICE:

Ocean pH has dropped from 8.2 to 8.1 since pre-industrial times. This 0.1 unit decrease represents approximately a 30% increase in hydrogen ion concentration because the pH scale is logarithmic. Based on your model, which consequence is MOST likely to appear first as pH continues to decline?

CONSTRUCTED RESPONSE:

Using your model, explain the chain of cause-and-effect relationships that connect a coal-fired power plant's CO_2 emissions to the collapse of an oyster fishery 3,000 miles away. Reference at least four model components and identify where tipping points exist in the chain.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Ocean acidification is often called climate change's 'equally evil twin' — a global chemical transformation happening beneath the waves that threatens marine ecosystems, fisheries worth $150 billion annually, and the ocean's fundamental ability to regulate Earth's climate. Understanding this crisis requires connecting atmospheric chemistry, carbonate geochemistry, marine biology, and complex systems dynamics.

The Chemistry of Ocean Acidification

When CO2 dissolves in seawater, it reacts with water molecules to form carbonic acid (H2CO3), which quickly dissociates into hydrogen ions (H+) and bicarbonate ions (HCO3-). The excess hydrogen ions then react with carbonate ions ($CO3^{2-}$) to form more bicarbonate. This double reaction simultaneously lowers pH and depletes the carbonate ions that marine organisms need for calcification. The ocean has absorbed approximately 525 billion tons of CO2 since 1800, lowering average surface pH from 8.2 to 8.1 — a 30% increase in acidity on the logarithmic pH scale. Current absorption rate is approximately 22 million tons of CO2 per day.

Biological Impacts Across Trophic Levels

Calcifying organisms are the most directly threatened: corals, mollusks, echinoderms, and pteropods all build structures from calcium carbonate that dissolves in acidified water. Pteropods — rice-grain-sized sea snails — are particularly critical because they form the base of polar food webs eaten by salmon, herring, and whales. Laboratory studies show pteropod shells dissolving within 45 days at pH levels projected for 2050. Coral reef calcification rates have declined 14% since 2000. Non-calcifying organisms are also affected: fish experience impaired sensory systems, altered behavior, and reduced reproductive success in acidified water.

Tipping Points and Irreversibility

Ocean acidification exhibits critical tipping points. Below a carbonate saturation state (Omega) of 1.0, seawater becomes corrosive to aragonite — the mineral form used by corals and pteropods. Models project that 70% of cold-water corals will be in undersaturated water by 2050. Once calcifying organisms decline past a critical population threshold, their role in the carbon cycle is lost — healthy shellfish beds actually help buffer pH by incorporating carbon into shells. The loss of this biological buffering capacity creates a positive feedback loop that accelerates acidification.

Economic and Food Security Dimensions

Global fisheries and aquaculture contribute $150 billion annually to the world economy and provide primary protein for over 3 billion people. The U.S. shellfish industry alone is worth $1.5 billion per year. Oregon and Washington oyster hatcheries have already experienced larval die-offs linked to acidified upwelling water, costing millions in losses. Coral reef ecosystems support tourism, coastal protection, and fisheries worth $375 billion per year globally. Communities in Southeast Asia, the Pacific Islands, and West Africa that depend on reef fisheries for food security face disproportionate risk from acidification.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify six components of the ocean acidification system: Atmospheric CO2 Concentration, Ocean CO2 Absorption Rate, Seawater pH, Carbonate Ion Availability, Shell-Building Organism Survival, and Marine Food Web Stability — distinguishing the external driver (atmospheric CO2) from internal system responses.

E **Establish Relationships**

Students determine that Atmospheric CO2 drives Ocean CO2 Absorption, which lowers Seawater pH, which depletes Carbonate Ion Availability, which threatens Shell-Building Organism Survival, which destabilizes Marine Food Web Stability — a cascading chain with critical tipping points at each link.

V **Visualize & Model**

Students build a computational model showing the cascade from atmospheric CO2 through ocean chemistry to ecosystem collapse, visualizing where nonlinear tipping points create sudden system shifts.

E **Evaluate Outcomes**

Students run pre-industrial, current trajectory, and aggressive mitigation scenarios to identify which CO2 pathways cross critical pH thresholds and whether marine ecosystems can recover once damage begins.

R **Revise & Extend**

Students add Ocean Temperature, Dissolved Oxygen, or Phytoplankton Productivity to explore how multiple simultaneous stressors interact synergistically with acidification.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The ocean has been quietly absorbing our carbon pollution for 200 years. It's been protecting us — but now it's paying the price. And the bill is coming due."*

Watch For

Show a time-lapse of pteropod shells dissolving in acidified water. Let students react. Ask: What could dissolve a shell?

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling a crisis most people have never heard of — one that's already reshaping life in the ocean."*

Watch For

Have students read objectives. Pre-teach pH scale and logarithmic meaning — emphasize that 0.1 pH change equals 30% more acid. Quick demonstration with pH paper and carbonated water.

Slide 3: Big Question

3 min

Say: *"We talk about CO2 in the atmosphere causing warming. But where does 25% of that CO2 actually go? Into the ocean. And what does it do there?"*

Watch For

Poll: Who has heard of ocean acidification? Compare awareness to global warming awareness. This is equally dangerous but far less known.

Slide 4: LEVER Framework

2 min

Say: *"We're modeling a chemical chain reaction that starts in a car exhaust pipe and ends with entire ecosystems dissolving."*

Watch For

Preview each LEVER step. Emphasize that this system has cascading tipping points — small changes accumulate until sudden collapse.

Slide 5: Activity 1: Components

8 min

Say: *"One external driver. Five internal responses. Where are the dominoes, and which one falls first?"*

Watch For

Guide sorting of external vs. internal components. Only Atmospheric CO_2 is truly external — everything else responds to the chemistry. Discuss why this makes the system vulnerable.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the chain: CO_2 enters the atmosphere, crosses into the ocean, changes the chemistry, dissolves shells, collapses food webs. Where can we break the chain?"*

Watch For

Help students map the cascade. Identify the critical nonlinear step: carbonate depletion accelerates sharply once pH drops below key thresholds. Find potential feedback loops.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's time-travel the ocean. First 1800, then 2100 on our current path, then 2100 if we act aggressively. Which future do you want?"*

Watch For

Students predict first, then run all three scenarios. Focus on when pH crosses the aragonite undersaturation threshold. Compare organism survival across scenarios.

Slide 8: Discoveries

5 min

Say: *"Your model just revealed why scientists call this the 'other CO_2 problem.' The ocean has been our silent ally — and we're destroying it."*

Watch For

Lead discussion on why ocean acidification receives less media attention than warming. Connect to real-world oyster hatchery failures in the Pacific Northwest. Discuss environmental justice for fishing communities.

Slide 9: STEM Challenge

12 min

Say: *"A marine sanctuary needs your expertise. Design a monitoring system that catches acidification before it kills the reef. Lives — and livelihoods — depend on it."*

Watch For

Groups design monitoring networks with sensor placement, threshold alerts, and intervention strategies. Present as conservation proposals.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Atmospheric CO2 Concentration
- **INTERNAL:** Ocean CO2 Absorption Rate, Seawater pH, Carbonate Ion Availability, Shell-Building Organism Survival, Marine Food Web Stability

Relationships

- **Atmospheric CO2 to Ocean CO2 Absorption Rate:** POSITIVE (+) — Higher Atmospheric CO2 increases the concentration gradient at the air-sea interface, driving more CO2 into ocean surface waters through gas exchange. The ocean absorbs approximately 25-30% of anthropogenic CO2 emissions.
- **Ocean CO2 Absorption Rate to Seawater pH:** NEGATIVE (-) — Higher CO2 absorption produces more carbonic acid, which dissociates into hydrogen ions and lowers pH. Each additional unit of CO2 absorbed drives pH further down on a logarithmic scale.
- **Seawater pH to Carbonate Ion Availability:** POSITIVE (+) — Lower pH means more hydrogen ions, which bind with carbonate ions to form bicarbonate. As pH drops, carbonate ion concentration decreases — at pH 7.8, carbonate levels drop below the threshold needed for aragonite shell formation.
- **Carbonate Ion Availability to Shell-Building Organism Survival:** POSITIVE (+) — Shell-building organisms require sufficient carbonate ions to construct and maintain calcium carbonate shells. When carbonate falls below saturation thresholds, shells dissolve faster than organisms can build them, leading to population collapse.
- **Shell-Building Organism Survival to Marine Food Web Stability:** POSITIVE (+) — Calcifying organisms form critical links in marine food webs — pteropods feed salmon, coral reefs shelter thousands of species, oyster reefs filter water. Their decline cascades through every trophic level above them.

Key Discoveries

- The ocean has absorbed roughly 30% of human CO2 emissions since the Industrial Revolution, making it approximately 30% more acidic — a rate of change unprecedented in at least 300 million years
- Carbonate ion availability drops exponentially as pH decreases, creating a tipping point where shell dissolution exceeds shell formation for most calcifying organisms
- Pteropods — tiny sea snails that form the base of many polar food webs — show shell dissolution at pH levels projected for 2050, threatening entire Arctic and Antarctic food chains
- Ocean acidification interacts synergistically with warming and deoxygenation, meaning the combined stress is far greater than any single stressor alone

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The ocean is becoming actual acid"	Despite the term 'acidification,' the ocean is not becoming acidic — it is becoming less alkaline. Ocean pH has dropped from 8.2 to 8.1, still on the basic side of the pH scale. The danger is the rate and direction of change, not the absolute pH value. Even small shifts in pH have enormous consequences for organisms adapted to narrow chemical ranges.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Ocean acidification only affects shellfish"	While calcifying organisms face the most direct chemical threat, acidification affects virtually all marine life. Fish exhibit impaired olfactory systems and altered behavior in acidified water, making them more vulnerable to predators. Phytoplankton communities shift composition, altering the base of the food web. Even organisms without shells are affected through food web disruption when their prey species decline.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The ocean can just keep absorbing CO_2 without consequences"	The ocean's capacity to absorb CO_2 is real but limited and comes with severe chemical side effects. As the ocean absorbs more CO_2, its buffering capacity decreases — meaning each additional ton of CO_2 causes a larger pH drop than the last. Furthermore, warming waters absorb less CO_2, potentially reducing the ocean's role as a carbon sink precisely when we need it most.